

**VIETNAM NATIONAL UNIVERSITY
UNIVERSITY OF ECONOMICS AND BUSINESS**

វិទ្យាបាល



FINAL REPORT

**REPORT: ADVENTUREWORK SALES ANALYSIS REPORT IN 2010 – 2013
PERIOD USING POWER BI**

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Semester I, Academic Year 2025-2026

ACKNOWLEDGMENT

The course Data Analysis in Economics and Business is a highly practical and valuable subject, equipping students with essential foundational knowledge and critical analytical skills that are closely aligned with real-world applications. My participation in this course has provided me with profound insights and valuable opportunities to hone my skills, thereby enhancing my professional competence in data processing and analysis.

I would like to express my deepest gratitude to Assoc. Prof., PhD. Vu Van Huong and PhD. Tran Thi Hoa Thom, the instructors who have directly taught and guided this course. The lecturers have devoted great passion to imparting knowledge and have created favorable conditions for students to practice with real-world datasets using Power BI. Their dedicated guidance and constructive feedback have been instrumental in helping me refine my skills and complete this report.

Once again, I extend my sincere thanks and best wishes to the lecturers for continued good health and further success in their educational and research careers.

Ha Noi, December 2025

Nguyen Thi Minh Anh

DECLARATION

I hereby solemnly declare that this report represents an original and independent research endeavor conducted entirely by me. The work has been carried out with the utmost academic integrity and honesty, without engaging in any form of plagiarism, unauthorized reproduction, or misappropriation of content from any external sources, publications, or prior research findings.

All data utilized in this report consist of secondary datasets that were provided by the course instructor and have been appropriately referenced in accordance with academic standards. These materials have been used exclusively for scholarly and research-oriented purposes within the scope of this academic exercise.

I fully acknowledge and accept complete responsibility for the content presented herein. Should any instance of academic misconduct—including but not limited to plagiarism, improper citation, or infringement of intellectual property rights—be identified, I undertake full accountability for such violations and any consequences that may arise thereof.

My sincere thanks!

Nguyen Thi Minh Anh

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1. INTRODUCTION

1.1. Project Context

In the contemporary business landscape, characterized by rapid advancements in information technology, organizations are increasingly leveraging technological tools to enhance operational efficiency and drive superior financial performance. Concurrently, the imperative to systematically collect, process, and analyze data has become a cornerstone of strategic and effective managerial decision-making. This is particularly salient for companies in sectors with intense competition and dynamic consumer trends. Enterprises that generate substantial transactional data throughout their value chain—especially from sales and marketing operations—face the critical challenge of transforming this raw data into actionable insights swiftly enough to respond to market fluctuations and formulate competitive strategies. Traditional, manual methods of analyzing large, complex, and perpetually updating datasets are often inefficient, potentially resulting in delayed strategic responses that can compromise a firm's market position and profitability.

In this environment, Business Intelligence (BI) and data visualization platforms, such as Microsoft Power BI, have emerged as transformative solutions to these challenges. These tools facilitate the efficient integration, processing, analysis, and visual representation of data, enabling organizations to move from descriptive reporting to diagnostic and predictive analytics. The application of Power BI allows businesses to critically assess the performance of past strategies, such as marketing campaigns and channel management, identify underlying trends, and subsequently make more informed, evidence-based decisions with greater speed. This study, therefore, serves as a practical exploration and demonstration of the significant utility of Power BI in supporting comprehensive data analysis and, consequently, enhancing the quality and effectiveness of business decision-making processes.

1.2. Research Objectives

The primary objective of this study is to construct a dynamic and interactive analytical dashboard that provides accurate, timely, and visually intuitive insights into the sales performance and operational strategies of AdventureWorks for the period

2010–2013. The research specifically focuses on analyzing key sales channels (Internet and Reseller), product category performance, customer demographics, and profitability trends. Based on the insights derived from this analysis, the study aims to evaluate the effectiveness of the company's dual-channel strategy and product lifecycle management, ultimately offering data-driven recommendations for strategic optimization.

The research delineates four specific and interconnected objectives. First, it aims to clean, transform, and prepare the historical sales dataset utilizing Power Query to establish a foundation of data integrity and analytical readiness. Following this preparatory phase, the study seeks to design and develop an integrated sales performance dashboard within the Power BI environment. This dashboard is intended to furnish both a holistic overview and granular, drill-down perspectives on AdventureWorks' sales dynamics, encompassing channel contributions, product mix performance, and regional market outcomes throughout the period from 2010 to 2013.

Building upon this analytical framework, the third objective is to undertake a critical evaluation of the effectiveness inherent in AdventureWorks' sales and multi-channel strategies over the specified four-year span. This evaluation will focus on identifying the principal drivers of growth, pinpointing episodes of significant volatility, and uncovering the underlying operational challenges that influenced performance. Ultimately, synthesizing the insights derived from this comprehensive analysis, the final objective is to formulate actionable strategic recommendations. These recommendations are directed towards enhancing future sales channel management, refining inventory planning processes, optimizing customer relationship strategies, and thereby improving the company's overall profitability trajectory.

1.3. Scope of Implementation

In terms of temporal and topical scope, this research is confined to analyzing the sales operations and strategic performance of AdventureWorks specifically during the years 2010 through 2013. The study utilizes a provided historical dataset pertaining to the company's sales transactions, which serves as the foundational source for all data processing, analysis, and visualization tasks. The implementation scope centers on the development of a comprehensive Power BI dashboard tailored to this dataset. The

analytical focus encompasses sales trend analysis, dual-channel (Internet vs. Reseller) performance evaluation, product category profitability assessment, and high-level customer segmentation analysis, all within the defined 2010-2013 timeframe.

2. DATA AND METHODOLOGY

2.1. Data Preprocessing

The initial step in the data preprocessing phase involves loading the data into Power Query. Upon uploading the data to Power Query, it is evident that the dataset comprises 15 tables, with two primary tables being Internet_Sales and Reseller_Sales. The supplementary tables contain related information essential for data analysis queries, such as Customer, Employees, Geography, Sales_Territory, Products, Product_Subcategory, Product_Category, among others. The processing of these tables will be conducted directly within Power Query. For this specific dataset, three tables require particular attention: Product_Subcategory, Product, and Internet_Sales_Reason.

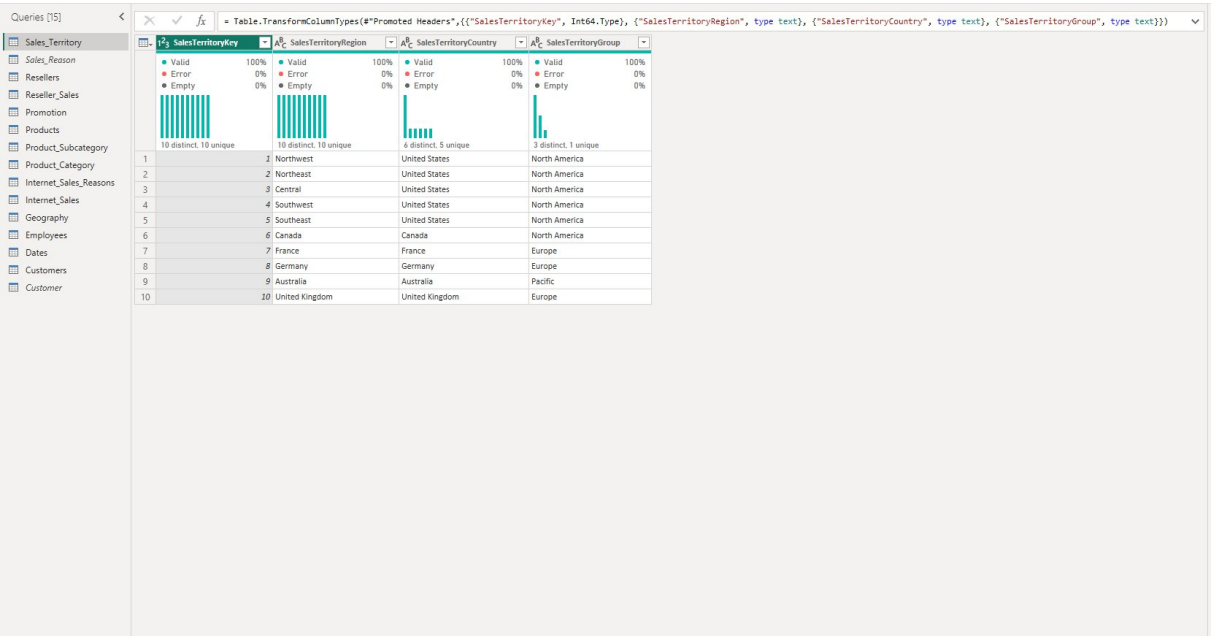


Figure 2.1. Loading Data in Power Query

2.2. Product Subcategory Table Process

Regarding the Subcategory table, the objective of processing this table is to integrate data from the Product_Category table into the Subcategory table. This consolidation aims to facilitate data queries and streamline the subsequent analytical process. To merge the data from the Product_Category table, within the Product_Subcategory table, we select Merge Query from the Combine group on the Home tab. The Merge Query dialog box will then appear, as illustrated in Figure 2.2.

Merge

Select a table and matching columns to create a merged table.

Product_Subcategory

ProductSubcategoryKey	SubcategoryName	ProductCategoryKey
1	Mountain Bikes	1
2	Road Bikes	1
3	Touring Bikes	1
4	Handlebars	2
5	Bottom Brackets	2

Product_Category

ProductCategoryKey	EnglishProductCategoryName
1	Bikes
2	Components
3	Clothing
4	Accessories

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

OK Cancel

Figure 2.2. Operation dialog in Merge Query for Product Subcategory Table

In this operation dialog, we select the two tables to be merged, which are the Product_Subcategory table and the Product_Category table. Subsequently, we choose the column that serves as the key for merging, which is the ProductCategoryKey column, and then click OK. After completing the merge query operation, we proceed to expand the newly merged column to include the desired content. In this instance, the content to be expanded from the Product_Category table is the EnglishProductCategoryName column.

×

Expand Product_Category

Select the columns to expand.

☒ (Select All Columns)
 ☐ ProductCategoryKey
 ☒ EnglishProductCategoryName

Default column name prefix (optional)

Product_Category

OK

Cancel

Figure 2.3. Expand value Table in Merge Query for Product Subcategory Table

Upon successfully expanding the content from the Product Category table, we rename the column to EnglishProductCategoryName. This completes the processing for the Product_Subcategory table. The finalized Product_Subcategory table is illustrated in Figure 2.4.

1 ² ProductSubcategoryKey	A ¹ SubcategoryName	1 ² ProductCategoryKey	A ¹ EnglishProductCategoryName
Valid 100% Error 0% Empty 0% 37 distinct, 37 unique	Valid 100% Error 0% Empty 0% 37 distinct, 37 unique	Valid 100% Error 0% Empty 0% 4 distinct, 0 unique	Valid 100% Error 0% Empty 0% 4 distinct, 0 unique
1	1 Mountain Bikes		1 Bikes
2	2 Road Bikes		1 Bikes
3	3 Touring Bikes		1 Bikes
4	4 Handlebars		2 Components
5	5 Bottom Brackets		2 Components
6	6 Brakes		2 Components
7	7 Chains		2 Components
8	8 Cranksets		2 Components
9	9 Derailleurs		2 Components
10	10 Forks		2 Components
11	11 Headsets		2 Components
12	12 Mountain Frames		2 Components
13	13 Pedals		2 Components
14	14 Road Frames		2 Components
15	15 Saddles		2 Components
16	16 Touring Frames		2 Components
17	17 Wheels		2 Components
18	18 Bib-Shorts		3 Clothing
19	19 Caps		3 Clothing
20	20 Gloves		3 Clothing
21	21 Jerseys		3 Clothing
22	22 Shorts		3 Clothing
23	23 Socks		3 Clothing
24	24 Tights		3 Clothing
25	25 Vests		3 Clothing
26	26 Bike Racks		4 Accessories
27	27 Bike Stands		4 Accessories
28	28 Bottles and Cages		4 Accessories
29	29 Cleaners		4 Accessories
30	30 Fenders		4 Accessories
31	31 Helmets		4 Accessories
32	32 Hydration Packs		4 Accessories

Figure 2.4. Product Subcategory Table after Merge Query

2.3. Product Table Process

For the Product table, the purpose of processing it is to merge data from the Product_Subcategory table into the Product table, thereby facilitating easier data querying during the analysis phase. To merge the data from the Product_Subcategory table, within the Product table, we select Merge Query from the Combine group on the Home tab. The Merge Query dialog box will then appear, as illustrated in Figure 2.5.

Merge

Select a table and matching columns to create a merged table.

Products

ProductKey	ProductSubcategoryKey	ProductName	StandardCost	FinishedGoodsFlag	Color
226	21	Long-Sleeve Logo Jersey, S	31.7244	TRUE	Multi
227	21	Long-Sleeve Logo Jersey, S	29.0807	TRUE	Multi
228	21	Long-Sleeve Logo Jersey, S	38.4923	TRUE	Multi
229	21	Long-Sleeve Logo Jersey, M	31.7244	TRUE	Multi

Product_Subcategory

ProductSubcategoryKey	SubcategoryName	ProductCategoryKey	EnglishProductCategoryName
1	Mountain Bikes	1	Bikes
2	Road Bikes	1	Bikes
3	Touring Bikes	1	Bikes
4	Handlebars	2	Components
5	Bottom Brackets	2	Components

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

OK Cancel

Figure 2.5. Operation dialog in Merge Query for Product Table

In this operation dialog, we select the two tables to be merged: the Product table and the Product_Subcategory table. Next, we choose the column that serves as the key for merging, which is the ProductSubcategoryKey column, and then click OK.

After completing the merge query operation, we proceed to expand the newly merged column to include the desired content. In this instance, the content to be expanded from the Product Subcategory table is the EnglishProductCategoryName column.

Expand Product_Subcategory

Select the columns to expand.

☒ (Select All Columns)
☐ ProductSubcategoryKey
☐ SubcategoryName
☐ ProductCategoryKey
☒ EnglishProductCategoryName

Default column name prefix (optional)

Product_Subcategory

OK Cancel

Figure 2.6. Expand value Table in Merge Query for Product Table

After successfully expanding the content from the Product Subcategory table, we rename the column to EnglishProductCategoryName, thereby completing the processing for the Product table. The finalized Product table is presented in Figure 2.7.

1 ² ₃ ProductKey	1 ² ₃ ProductSubcategoryKey	A _C ProductName	1.2 StandardCost	FinishedGoodsFlag	A _C Color	StartDate	A _C ProductCategoryName
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
395 distinct, 395 unique	37 distinct, 8 unique	293 distinct, 216 unique	134 distinct, 52 unique	1 distinct, 0 unique	10 distinct, 1 unique	3 distinct, 0 unique	4 distinct, 0 unique
226	21	Long-Sleeve Logo Jersey, S	31.7244	TRUE	Multi	01-Jul-11 12:00:00 AM	Clothing
227	21	Long-Sleeve Logo Jersey, S	29.0807	TRUE	Multi	01-Jul-12 12:00:00 AM	Clothing
228	21	Long-Sleeve Logo Jersey, S	38.4923	TRUE	Multi	01-Jul-13 12:00:00 AM	Clothing
229	21	Long-Sleeve Logo Jersey, M	31.7244	TRUE	Multi	01-Jul-11 12:00:00 AM	Clothing
230	21	Long-Sleeve Logo Jersey, M	29.0807	TRUE	Multi	01-Jul-12 12:00:00 AM	Clothing
231	21	Long-Sleeve Logo Jersey, M	38.4923	TRUE	Multi	01-Jul-13 12:00:00 AM	Clothing
232	21	Long-Sleeve Logo Jersey, L	31.7244	TRUE	Multi	01-Jul-11 12:00:00 AM	Clothing
233	21	Long-Sleeve Logo Jersey, L	29.0807	TRUE	Multi	01-Jul-12 12:00:00 AM	Clothing
234	21	Long-Sleeve Logo Jersey, L	38.4923	TRUE	Multi	01-Jul-13 12:00:00 AM	Clothing
235	21	Long-Sleeve Logo Jersey, XL	31.7244	TRUE	Multi	01-Jul-11 12:00:00 AM	Clothing
236	21	Long-Sleeve Logo Jersey, XL	29.0807	TRUE	Multi	01-Jul-12 12:00:00 AM	Clothing
237	21	Long-Sleeve Logo Jersey, XL	38.4923	TRUE	Multi	01-Jul-13 12:00:00 AM	Clothing
397	4	LL Mountain Handlebars	17.978	TRUE	NA	01-Jul-12 12:00:00 AM	Components
398	4	LL Mountain Handlebars	19.7758	TRUE	NA	01-Jul-13 12:00:00 AM	Components
391	10	LL Fork	65.8097	TRUE	NA	01-Jul-12 12:00:00 AM	Components
392	10	ML Fork	77.9176	TRUE	NA	01-Jul-12 12:00:00 AM	Components
393	10	HL Fork	101.8936	TRUE	NA	01-Jul-12 12:00:00 AM	Components
410	17	LL Mountain Front Wheel	26.9708	TRUE	Black	01-Jul-12 12:00:00 AM	Components
411	17	ML Mountain Front Wheel	92.8071	TRUE	Black	01-Jul-12 12:00:00 AM	Components
412	17	HL Mountain Front Wheel	133.2955	TRUE	Black	01-Jul-12 12:00:00 AM	Components
416	17	Touring Front Wheel	96.7964	TRUE	Black	01-Jul-12 12:00:00 AM	Components
419	17	LL Mountain Rear Wheel	38.9588	TRUE	Black	01-Jul-12 12:00:00 AM	Components
420	17	ML Mountain Rear Wheel	104.7951	TRUE	Black	01-Jul-12 12:00:00 AM	Components
421	17	HL Mountain Rear Wheel	145.2835	TRUE	Black	01-Jul-12 12:00:00 AM	Components
425	17	Touring Rear Wheel	108.7844	TRUE	Black	01-Jul-12 12:00:00 AM	Components
445	22	Men's Sports Shorts, S	24.7459	TRUE	Black	01-Jul-12 12:00:00 AM	Clothing
453	22	Men's Sports Shorts, M	24.7459	TRUE	Black	01-Jul-12 12:00:00 AM	Clothing
454	22	Men's Sports Shorts, L	24.7459	TRUE	Black	01-Jul-12 12:00:00 AM	Clothing
455	22	Men's Sports Shorts, XL	24.7459	TRUE	Black	01-Jul-12 12:00:00 AM	Clothing
474	22	Women's Mountain Shorts, S	26.1763	TRUE	Black	01-Jul-13 12:00:00 AM	Clothing
475	22	Women's Mountain Shorts, M	26.1763	TRUE	Black	01-Jul-13 12:00:00 AM	Clothing
476	22	Women's Mountain Shorts, L	26.1763	TRUE	Black	01-Jul-13 12:00:00 AM	Clothing

Figure 2.7. Product Table after Merge Query

2.4. Internet_Sales_Reason Table Process

For the Internet_Sales_Reason table, it is necessary to remove any duplicate entries in the SaleOrderNumber column to ensure data uniqueness, meaning each SaleOrderNumber must be distinct.

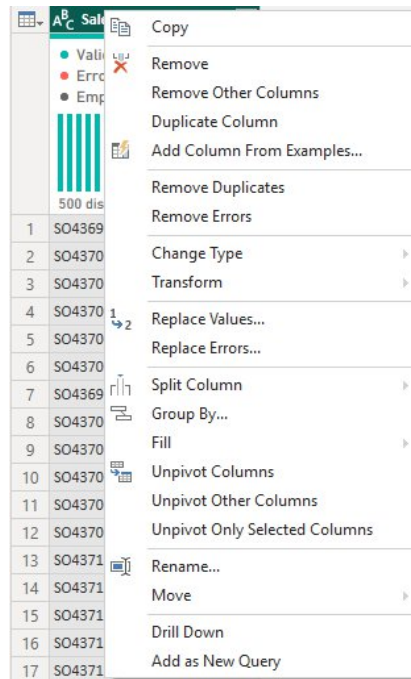


Figure 2.7. Remove duplicate for Internet Sales Reason Table

2.5. Table Relationship Management

To ensure accurate inter-table connectivity, it is essential to establish appropriate relationships among the tables. Figures 2.8 and 2.9 illustrate the relationship management interface, depicting the adjusted and finalized linkages between the tables.

Manage relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐	From: table (column) ↑	Relationship	To: table (column)	Status	
☐	Customers (GeographyKey)		Geography (GeographyKey)	Active	...
☐	Employees (SalesTerritoryKey)		Sales_Territory (SalesTerritoryK...	Inactive	...
☐	Geography (SalesTerritoryKey)		Sales_Territory (SalesTerritoryK...	Inactive	...
☐	Internet_Sales (CustomerKey)		Customers (CustomerKey)	Active	...
☐	Internet_Sales (OrderDate)		Dates (Date)	Active	...
☐	Internet_Sales (ProductKey)		Products (ProductKey)	Active	...
☐	Internet_Sales (PromotionKey)		Promotion (PromotionKey)	Active	...
☐	Internet_Sales (SalesOrderNu...		Internet_Sales_Reasons (Sales...	Active	...
☐	Internet_Sales (SalesTerritoryK...		Sales_Territory (SalesTerritoryK...	Active	...
☐	Product_Subcategory (Product...		Product_Category (ProductCat...	Active	...
☐	Products (ProductSubcategory...		Product_Subcategory (Product...	Active	...
☐	Reseller_Sales (EmployeeKey)		Employees (EmployeeKey)	Active	...

Close

Figure 2.8. Relationship Management

Manage relationships



+ New relationship		Autodetect	Edit		Delete	Filter
☐ From: table (column) ↑	Relationship	To: table (column)	Status			
☐ Internet_Sales (SalesTerritoryK...	* — 1	Sales_Territory (SalesTerritoryK...	Active	...		
☐ Product_Subcategory (Product...	* — 1	Product_Category (ProductCat...	Active	...		
☐ Products (ProductSubcategory...	* — 1	Product_Subcategory (Product...	Active	...		
☐ Reseller_Sales (EmployeeKey)	* — 1	Employees (EmployeeKey)	Active	...		
☐ Reseller_Sales (OrderDate)	* — 1	Dates (Date)	Active	...		
☐ Reseller_Sales (ProductKey)	* — 1	Products (ProductKey)	Active	...		
☐ Reseller_Sales (PromotionKey)	* — 1	Promotion (PromotionKey)	Active	...		
☐ Reseller_Sales (ResellerKey)	* — 1	Resellers (ResellerKey)	Active	...		
☐ Reseller_Sales (SalesTerritoryK...	* — 1	Sales_Territory (SalesTerritoryK...	Active	...		
☐ Resellers (GeographyKey)	* — 1	Geography (GeographyKey)	Inactive	...		

Close

Figure 2.9. Relationship Management (cont)

3. DATA ANALYSIS

3.1. Revenue Development Trend



Figure 3.1. Development trends analysis in the 2010–2013 period

In the 2010–2013 period, the year 2010 marked a foundational beginning for AdventureWorks, not merely as the launch of a store or the establishment of a website, but as a process of shaping a data-driven management philosophy. With total revenue remaining modest at only 10.8 million and profits around 1.61 million, at this point the enterprise had begun operating a "dual-channel" model, embarking on a calculated market penetration strategy. The company chose the Reseller channel in North America as its initial market, viewing it as a brand "anchor," serving as a basis for future market entry. The clearest indicator of this is that, although the reseller channel had only just become active in North America this year, revenue from this region accounted for 70.16% of AdventureWorks' total annual revenue. Although transaction volume was not yet large, establishing relationships with strategic resellers helped the company understand logistics cost structures and cost of goods sold (COGS), laying the groundwork for process optimization in the next phase. Conversely, although the Internet channel operated in all three regions—North America, Pacific, and Europe—

its revenue contribution accounted for only 29.84%. Overall, AdventureWorks' total revenue in 2010 was not yet high, but it can be seen as a pivotal year for establishing logistics processes and identifying key target markets.

On the other hand, sales indicators also revealed the development potential of the Internet channel, as the profit from this channel accounted for up to 80.58% of the year's total profit. The reason for this lies in the channel's online advantages, enabling optimization of product-related costs and reducing COGS to the lowest level, accounting for only 20.98% of total COGS, while selling prices did not differ significantly from the Reseller channel. This resulted in a clear superiority in the Internet channel's profitability. Finally, although still small in scale, the emergence of the "dual-channel" model helped the enterprise recognize core differences in customer behavior: while resellers typically placed large bulk orders mid-week, retail customers maintained a more spread-out purchasing rhythm, creating a perfect complement in terms of cash flow from the very beginning of operations.

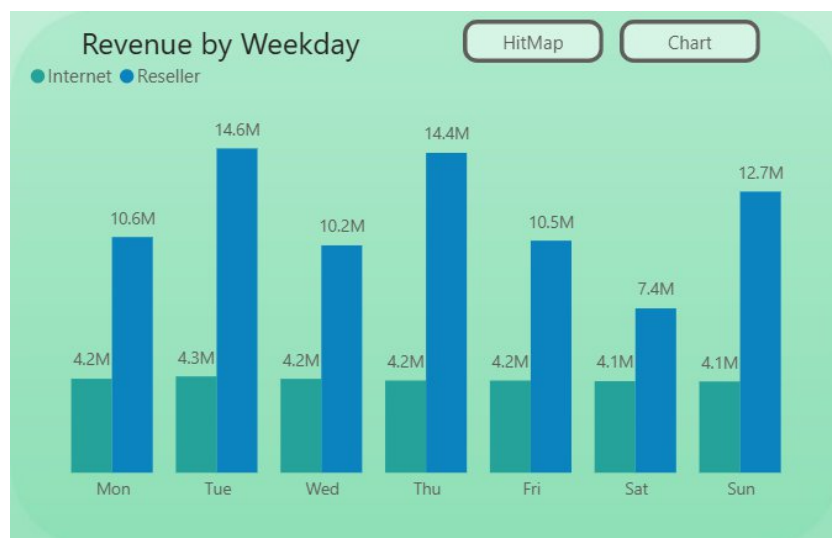


Figure 3.2. Revenue by Weekday

Subsequently, moving into 2011, AdventureWorks witnessed a "rapid" growth spurt as revenue surged to 32.54 million USD, a threefold increase compared to the previous year, demonstrating the explosion in its distribution network scale. This was the year of breakthroughs in wholesale scale, a "hot growth" phase with the Reseller

channel acting as the main engine. A notable example was the surge in August 2011, where the reseller channel's month-over-month (MoM) growth reached a record 371%, showcasing the company's ability to expand its distribution network. However, accompanying this rapid growth was the instability of the Reseller channel, as the profit chart showed sharp, sawtooth-like fluctuations. This suggests that sales channels through resellers are extremely sensitive to various influencing factors such as COGS, reseller discount programs, or environmental and customer factors. The reasons for this instability could also stem from internal factors like risk management and cash flow stability. This required the enterprise to begin focusing more on controlling COGS and related programs to protect net profit amidst increasing operational and competitive pressures. Concurrently, the Internet Sales channel began to demonstrate its role as a "stable revenue source" with steady month-to-month growth, helping the business reduce financial pressure from fluctuations in the wholesale market.

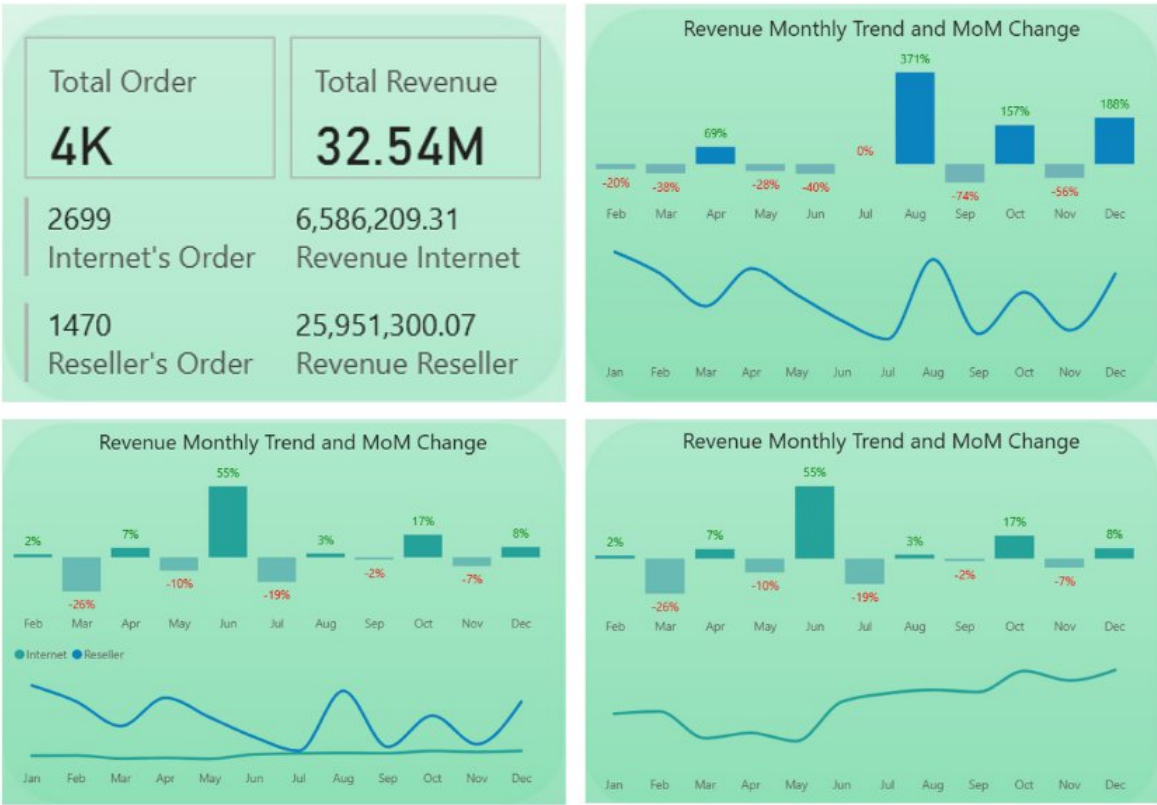


Figure 3.3. Revenue development in 2012

However, the pinnacle of this entire development cycle lies in 2012, the "golden year" with a record revenue of 42.35 million USD, where operational optimization reached maturity. During this phase, AdventureWorks expanded not only in quantity but also achieved operational efficiency maturity. Revenue from the Reseller channel accounted for 76.85% of the year's total revenue, with resellers in North America holding the highest proportion at 26 million in revenue, concentrated most in the Southwest region of the United States (7.3 million revenue) and Canada (5.6 million revenue). Furthermore, the customer group in Australia was the most profitable, contributing approximately 1.25 million, accounting for about 32.28% of the total profit for that year. Notably, the majority of profit from this customer group originated from the Internet channel (approximately 0.89 million USD). This demonstrates that the geographical diversification strategy had become an effective safety net, as the explosive growth of the Internet channel in the Australian market compensated for the cyclical fluctuations of the Reseller channel in Europe and America. Most importantly, the combination of the massive scale of the Reseller channel in North America and the high profit margins from this Australian customer group created an optimal financial structure. This was also the year the enterprise achieved Economies of Scale, evident in its leveraging of scale advantages to reduce marginal costs per unit product, creating a significant gap between revenue and cost of goods sold (COGS), resulting in the widest profit margin in the four-year period from 2010 to 2013.



Internet

Reseller

6

[--]

(\$)Total Profit by Region

Region	Profit
Australia	0.89M
Southwest	0.52M
Northwest	0.34M
Canada	0.25M
United Kingdom	0.24M
Germany	0.21M
Central	
Northeast	
Southeast	

2,458,556 out of 2,669,681 Internet Profit

Internet

Reseller

6

92.09% of Internet's Total Profit is come from Top-6 region

(\$)Total Profit by Region in Internet Channel

Region	Profit
Australia	0.89M
Southwest	0.52M
Northwest	0.34M
Canada	0.25M
United Kingdom	0.24M
Germany	0.21M

2,458,556 out of 2,669,681 Internet Profit

Figure 3.4. Revenue development in 2013

The year 2013 concluded the cycle with a multifaceted picture, reflecting a shift from quantity to quality. Although aggregate revenue data showed signs of decline (reaching only 24.12 million USD), this decline primarily stemmed from the restructuring of the Reseller channel, while the Internet channel maintained its performance as a reliable cash cow, sustaining revenue levels similar to the previous year, proving its enduring vitality. The most significant difference this year was the change in reseller purchasing habits. While the Internet channel remained consistent and stable across all days of the week, Reseller channel revenue showed a clear shift, beginning to concentrate on weekdays from Tuesday to Friday and on Sundays. Conversely, Mondays and Saturdays saw almost no orders. This is an indicator of a change in the global market supply cycle. This forced the enterprise to plan a restructuring of its operational schedule, optimizing human resources and logistics, and

shifting resources according to specific timeframes to ensure work efficiency remained at its peak without incurring excess management costs. The harmonious coordination between the two channels based on timeframes not only optimized manpower but also ensured that working capital flowed most efficiently, guaranteeing liquidity for the entire corporate system amidst its continuously expanding scale.

Thus, overall, 2013 should not be viewed as a recession, but rather as a "self-adjustment" phase to optimize profit margins and refine the reseller network, preparing for a more sustainable business cycle.

In summary, the development phase of AdventureWorks from 2010–2013 highlighted key aspects of its business strategy. The most prominent feature is the operational strategy based on an "interconnected vessel" model between the two opposing distribution channels. The combination of Internet Sales, a stable revenue source with high-profit margins, and Reseller Sales, a channel generating explosive sales volume on a massive scale, enabled the enterprise to build a stable financial ecosystem. These two channels do not exist independently but consistently complement each other in terms of cash flow. When the wholesale segment faces market fluctuations or seasonal impacts, the stable growth from online retail acts as a "safety net," helping the enterprise mitigate systemic risks and maintain continuous development momentum.

Parallel to optimizing the channel structure, it is evident that due to relatively stable and fiercely competitive product market prices, the key to increasing surplus value lies in the ability to tightly control the Cost of Goods Sold (COGS). Analytical data across the years shows that any slight reduction in operational or sales costs creates a significantly larger positive impact on net profit compared to efforts to increase selling prices.

Furthermore, the analysis of AdventureWorks' development trends reveals the enterprise's acumen in establishing a flexible logistics ecosystem based on the specific purchasing habits of each channel. Accurately identifying the "golden days" of the week—such as concentrating on processing wholesale orders mid-week while

maintaining flexible retail delivery services throughout the week—allowed AdventureWorks to personalize services while maximizing cost savings. This approach not only enhanced the experience for both reseller and retail customers but also ensured the most efficient circulation of working capital, maintaining liquidity for the entire corporate system throughout the business cycle.

However, to operate most effectively, based on the analyzed data, AdventureWorks still has unresolved issues in the data that need clarification:

First, why did revenue experience such remarkable growth from 2010 to 2011 and peak in 2012? Was it due to changes in product structure or promotional programs?

Second, upon examining detailed monthly data across the years, a phase shift between Quantity, Revenue, and Gross Profit can be observed sporadically throughout all four years, but it is particularly evident in the last three months of 2011. The question here is why is there such a clear discrepancy or contradiction between these sales indicators? Could it be due to changes in product mix or another reason?

Total Quantity 3K	Total Revenue 2.98M	Total COGs 3M	Total Gross Profit 372.05K	% Profit Margin 12.50%
Sep : 1,445 Variance :▲+120.5%	Sep : 1,485,983 Variance :▲+100.4%	Sep : 1,214,833 Variance :▲+114.5%	Sep : 271,150 Variance :▲+37.2%	Sep : ▲+18.25%
Total Quantity 2K	Total Revenue 1.66M	Total COGs 2M	Total Gross Profit -338.28K	% Profit Margin -20.35%
Oct : 3,186 Variance :▼-24.3%	Oct : 2,977,325 Variance :▼-44.2%	Oct : 2,605,279 Variance :▼-23.2%	Oct : 372,046 Variance :▼-190.9%	Oct : ▲+12.5%
Total Quantity 9K	Total Revenue 3.60M	Total COGs 3M	Total Gross Profit 360.89K	% Profit Margin 10.04%
Nov : 2,412 Variance :▲+254.8%	Nov : 1,662,350 Variance :▲+116.3%	Nov : 2,000,626 Variance :▲+61.7%	Nov : -338,276 Variance :▲+206.7%	Nov : ▼-20.35%

Figure 3.5. Sales performance metrics across October, November, and December of 2011

Third, there is the sudden "disappearance" of reseller channel revenue. Data from 2013 shows that in Q1 and Q2 of 2013 (i.e., Q3 and Q4 of fiscal year 2012), almost no reseller channel revenue was reported. In the latter half of 2013, the reseller channel began to record revenue again. However, compared to the stable growth of the Internet channel, the reseller channel witnessed a clear decline in revenue from Q3 to Q4 (i.e., Q1 and Q2 of fiscal year 2013). Thus, the question arises: What is the cause of this decline? Is it due to a market freeze or did we stop supplying a key product line?



Figure 3.6. The revenue trend throughout the year 2013

To answer these two pivotal questions, we will proceed with a detailed analysis of each sales channel: Internet Sales and Reseller Sales.

3.2. Internet Sales Analysis

Regarding the Internet sales channel, in the first year of the 2010-2013 period, the Internet Sales channel operated as a "premium laboratory" with an extremely streamlined product portfolio, focusing 100% on complete Bikes such as Road Bikes and Mountain Bikes. This focus on a single product category indicates that AdventureWorks was aiming for a "premium brand positioning" in the digital space in 2010. In fact, the enterprise was quite successful in this strategy, as most buyers chose to purchase due to the manufacturer's brand, which was the purchasing reason that generated \$2.6 million USD.

The enterprise did not sell individual components retail but concentrated on selling complete bicycles with the highest value to establish credibility before expanding into more mass-market segments. This strategy helped the enterprise build brand credibility in markets with existing infrastructure, such as North America and Australia,

before considering expansion. These were also the two regions contributing the most revenue in 2010, with total revenue from the top three regions amounting to approximately \$2.4 million USD. On the other hand, the main subcategories sold under Bikes were Road Bikes with \$2.7 million in revenue and Mountain Bikes with \$0.6 million in revenue, both of which are products with extremely high unit prices. This explains why the Total Quantity was very low at 999 orders in 2010, yet the total revenue for 2010 reached \$3.22 million USD.

Furthermore, although each transaction generated such a high average order value, the actual profit realized was only \$1 million USD. This also helped the enterprise gain a deep understanding of the cost structure for Cost of Goods Sold (COGS) and logistics for these bulky items. Revenue analysis by Category shows that although Road Bikes generated high revenue, their profit margin was relatively low, at only about 39.37%. Conversely, Mountain Bikes had lower revenue but a significantly higher profit margin, approximately 45%. Recognizing this could be one reason leading to the freeze in the latter half of 2010 when no revenue was recorded. However, the cause of this freeze could also stem from other situations, such as system restructuring, warehouse upgrades, or website platform changes in preparation for the next phase.

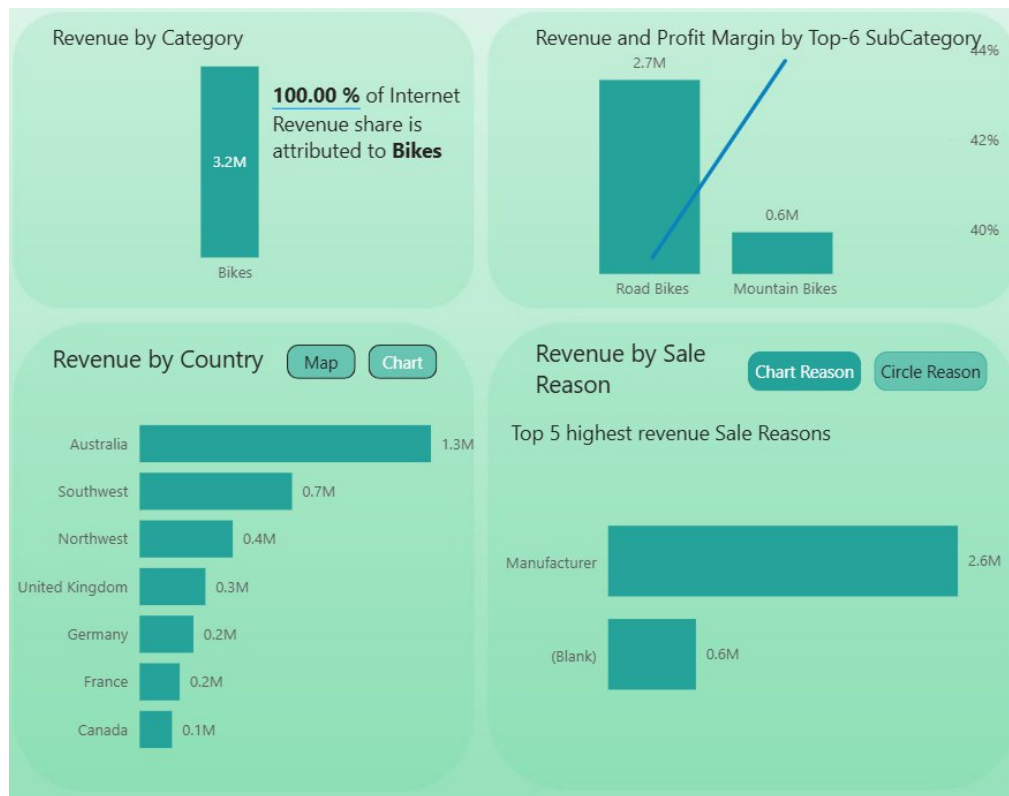


Figure 3.7. Revenue analysis of Internet Sales in 2010

Moving into 2011, this year marked a significant turning point in AdventureWorks' product strategy, as the Mountain Bikes category gradually gained a more important position on the Internet channel, sharing the dominance held by Road Bikes in 2010. This shift became most evident in the revenue figures for the first half of 2011, showing steady growth and peaking in June 2011 with \$230,000 USD, accounting for 41.07% of total revenue for that month. During this period, the company aggressively promoted the Mountain-200 and Mountain-100 series, creating a surge that more than doubled revenue compared to the previous year. Concurrently, with their high-profit margins, Mountain Bikes generated \$710,000 USD in profit, accounting for nearly one-third of the total profit in 2011 with only 621 orders—approximately 30% of the order volume of Road Bikes. This represents not only a change in product offerings but also a calculated strategy to expand the customer base, helping the brand reach consumers who enjoy exploring rugged terrain, moving beyond the initial focus solely on the professional cycling segment.

In this year, Australia remained the leading region in revenue with \$2.2 million USD, far surpassing the second-ranked region, the Southwest United States. Leading in revenue for two consecutive years, Australia proved itself to be an excellent market for AdventureWorks' Internet channel to exploit. Furthermore, analysis of customer purchasing reasons showed that another driver for increased order volume in 2011 was the introduction of new promotional programs and enhanced marketing efforts to attract online buyers. These two types of activities attracted approximately 49.1% of total revenue in 2011.

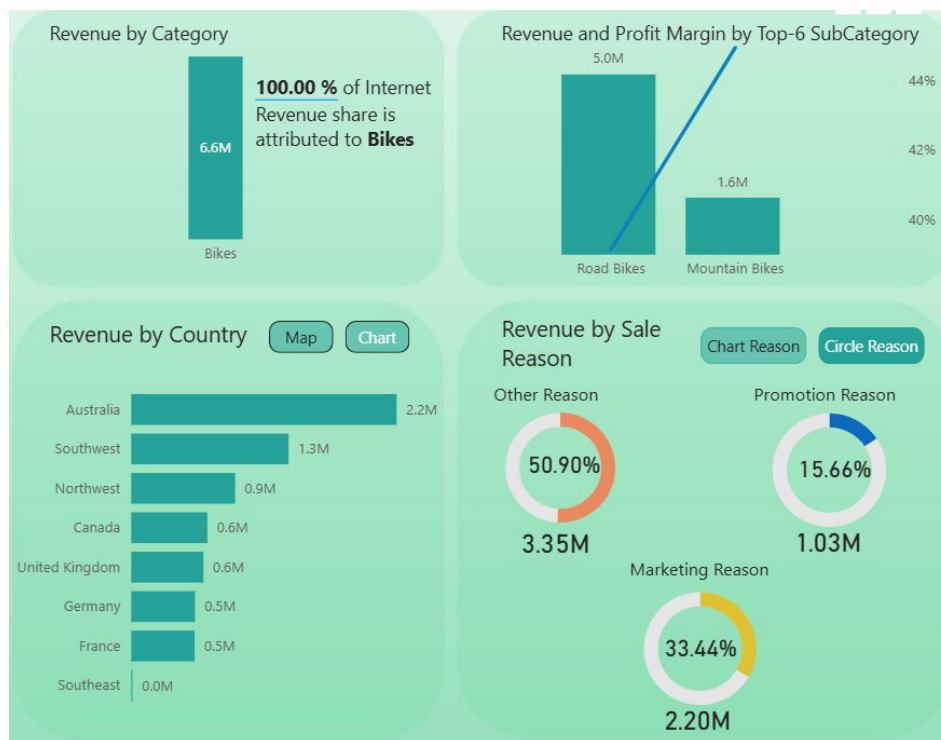


Figure 3.8. Revenue analysis of Internet Sales in 2011

The year 2012 was the peak revenue year for AdventureWorks in the 2010–2013 period. It also marked the beginning of a promising new chapter as AdventureWorks officially transitioned to a "Multi-category Retail" model with the introduction of Accessories and Clothing categories. This change created a fascinating business structure, manifested in a positive phase shift between revenue and quantity. While Bikes remained the "backbone," contributing over 90% of total revenue at approximately \$9.37 million USD, the Accessories category came to dominate over

70% of total order volume, with 19,000 items purchased. This indicates that introducing low-value products such as water bottles, helmets, or tires allowed the company to thoroughly solve the challenge of customer retention. Instead of waiting for the multi-year bicycle replacement cycle, AdventureWorks could now attract customers to make repeat purchases monthly, creating an extremely effective strategy to increase purchase frequency.

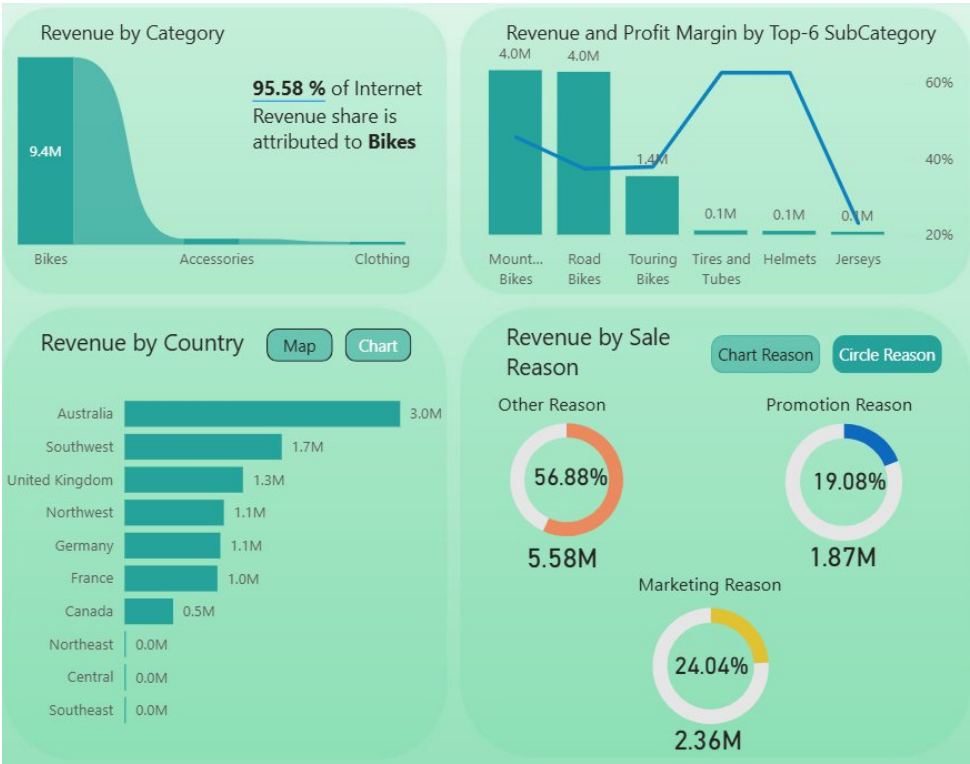


Figure 3.9. Revenue analysis of Internet Sales in 2012

This very diversification helps explain the phenomenon of explosive scale in the first six months of 2012, a period recording a highly vibrant financial picture. June 2012 became the breakout point, recording the second-highest number of orders in the entire 2010-2013 period for the company, with 5,000 orders. Although revenue did not far exceed other peak months, the number of shipments was the highest in 2012. More importantly, the net profit margin during this period remained relatively well-maintained and stable despite the addition of new product types. This demonstrates that AdventureWorks had found its "sweet spot" in business: using Bikes as the revenue-driving product and leveraging the Accessories group—with extremely low COGS and

cheap packaging costs—to protect and optimize profits, perfectly compensating for incurred shipping costs.

Furthermore, in 2012, Australia continued to be AdventureWorks' leading revenue region, not only demonstrating the most outstanding product adaptability within the system but also confirming its superior market potential. Customers in this market not only favored the company's bicycle lines but also had the highest rate of purchasing complementary accessories compared to other regions like the US or Europe. The success in Australia in 2012 sent a strong strategic message: the online retail model has the full potential to replace the role of physical resellers in markets with vast geography. Purchasing small accessories online was preferred due to its convenience and cost-effectiveness compared to customers having to travel to traditional stores, opening a new direction for optimizing the global distribution network.

However, from July to November 2012, AdventureWorks' Internet channel abruptly stopped selling both supplementary product groups, Accessories and Clothing. Based on prior data, this phenomenon could be a necessary reaction to year-end logistics overload pressure. In the first half of 2012, the explosion of small accessory orders placed immense strain on the online order processing system. Upon entering the year-end peak season for bicycle purchases, especially in the Australian market entering summer, the company likely proactively paused the sale of small retail items on the Internet channel. This allowed it to concentrate all warehousing, packaging, and shipping resources on bicycles—products of higher value and more complex logistics. Prioritizing "Bikes First" ensured the timely delivery of high-value orders during the holiday season, preventing small accessory items from clogging the quality control pipeline for premium bicycle lines.

Moreover, selling only Bikes during this period might also reflect a "Large-Scale Inventory Clearance" strategy. After successfully testing accessories in the first half of the year, AdventureWorks may have shifted all remaining accessory inventory to the Reseller channel for direct distribution. This aimed to clear out central warehouse space for receiving new 2013 bicycle models. This explains why Bikes revenue in the second

half of 2012 was extremely stable and overwhelmingly dominant, as all online marketing efforts were focused on clearing bicycle inventory before the fiscal year-end.

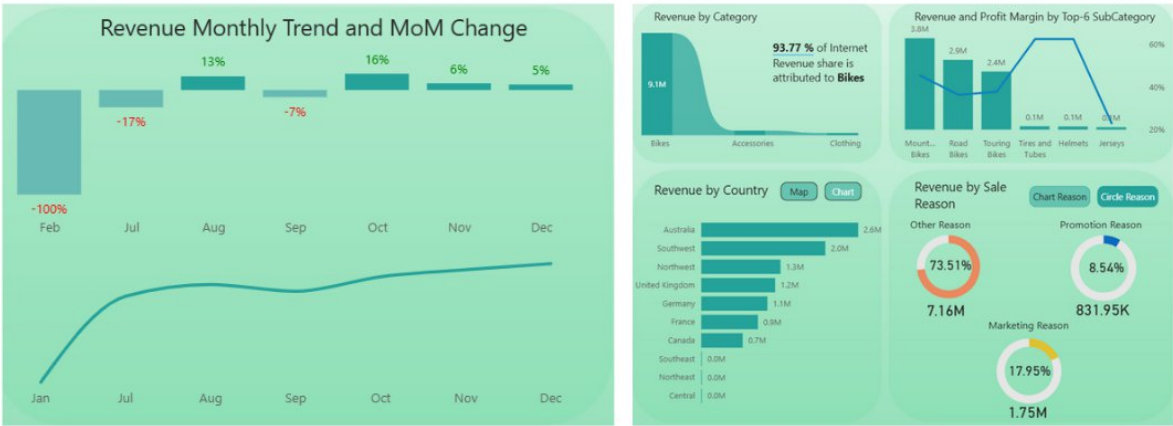


Figure 3.10. Revenue analysis of Internet Sales in 2013

The year 2013 began with a highly volatile data scenario, recording prolonged revenue gaps from February to June, with the sole bright spot in January. This unusual "hiatus" does not reflect a declining market but is essentially a signal of a large-scale internal transformation, commonly seen when a company undertakes a comprehensive e-commerce platform upgrade or an ERP system migration. Evidence lies in the stable January 2013 revenue from accessories and clothing, indicating robust consumer demand. However, AdventureWorks' fulfillment capability was temporarily disrupted by technical and operational factors, preparing for a longer-term business strategy.



Figure 3.11. Revenue analysis of Internet Sale in January 2013

The fact that in January 2013 the company sold only two product types, with no bicycles, could stem from having focused on selling out this category in the latter half of 2012. If the "Large-Scale Inventory Clearance" hypothesis for late 2012 holds true,

it suggests AdventureWorks went all-in to close the fiscal year. Conversely, this would have led to the bicycle inventory being completely depleted by early January 2013. Therefore, to maintain its online presence and utilize existing operational infrastructure, AdventureWorks pushed the remaining items in stock—Accessories and Clothing—onto the Internet platform. This was not a deliberate long-term strategy but a flexible response to optimize resources, selling what remained "leftover" from the massive bicycle clearance campaign.

When official data resumed in the last six months of 2013, a completely new product structure had clearly formed, marking the emergence of a "Long-tail Strategy." At this point, while sales of core bicycle lines (Bikes) showed signs of plateauing, categories like Accessories and Clothing—such as Tires and Tubes, Helmets, or Jerseys—reached their highest Total Quantity in the entire period, with over 23,000 orders. This is the stage where AdventureWorks broke away from dependence on new bicycle buyers and began fully exploiting "Customer Lifetime Value" (LTV). Users who purchased bikes during 2011-2012 now became the primary revenue source through repeat purchases of replacement parts and specialized apparel, creating a sustainable consumption ecosystem.

This shift led to a remarkable "counter-current" surge in profit in Q4 2013, creating a scenario completely opposite to the severe decline in the wholesale (Reseller) channel. While resellers cut back on inventory purchases due to a saturated complete-bike market, the Internet Sales channel reached revenue peaks in November and December thanks to the explosion of small retail orders. This confirms that Internet Sales had become the financial "lifesaver," helping AdventureWorks not only withstand the stagnation of the traditional channel but also find new growth momentum from direct retail of auxiliary parts to end consumers.

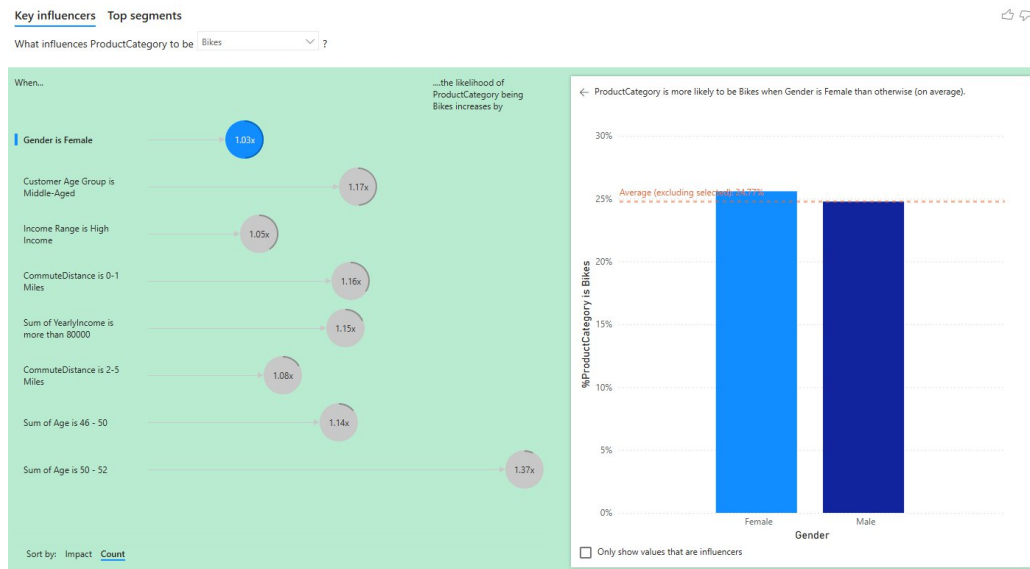


Figure 3.12. Internet's Customer analysis for Bikes



Figure 3.13. Internet's Customer analysis for Accessories

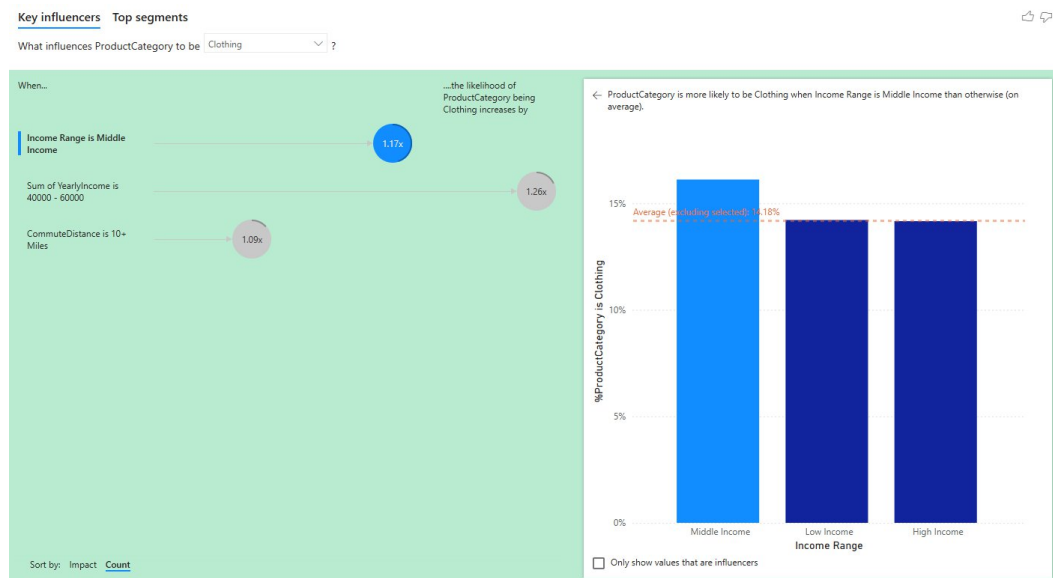


Figure 3.14. Internet's Customer analysis for Clothing

Furthermore, beyond sales metrics, the data also sketches portraits of customer groups on the Internet sales platform. Specifically, customers who tend to buy more Bikes are typically in a "major investment" phase. Most buyers of this product are concentrated in the age group of 49 to 69, with those aged 49 to 52 purchasing Bikes at a rate about 1.4 times the average. This age group has stable income and a growing interest in health and professional experiences. Another characteristic is that bicycle buyers often have high incomes, typically above \$80,000 USD annually, providing substantial discretionary budget for expensive hobbies. Next, customers who buy Accessories are usually in the age range of 65–82, a demographic that highly prioritizes health maintenance and tends to choose healthy activities like cycling. Also, customers who frequently commute distances from 5 to over 10 miles purchase accessories at a rate 1.06 to 1.08 times the average. Finally, customers who buy Clothing are a younger group, typically adults under 45, with average annual incomes of \$40,000 - \$60,000 USD or below, and often place greater emphasis on appearance and aesthetics.

Thus, the customer portrait for AdventureWorks on the Internet channel is a blend of professionalism, practicality, and lifestyle. Clearly identifying these customer characteristics will enable the company to execute highly precise targeted marketing strategies.

3.3. Reseller Sales Analysis

Regarding the Reseller sales channel, in the first year of the 2010–2013 period, this channel exhibited a "discontinuous" data pattern, with revenue data only present for January, March, and May of 2010. All other months recorded no transaction data. However, based on related information, this was not a system error but rather clear evidence of an intentional "Batch Sales" strategy. In the initial phase, AdventureWorks likely adopted a Seasonal Wholesale model, executing large "bulk shipments" by quarter instead of continuous sales. This approach created "breathing cycles" for the supply chain, allowing resellers sufficient time to distribute and sell the massive inventory before receiving new shipments. Notably, the first half of 2010 was also when the Internet Sales channel was in its pilot operation phase, indicating the company wisely used the direct channel as a "market gauge" to probe real user demand and subsequently adjust supply plans for the next wholesale push.

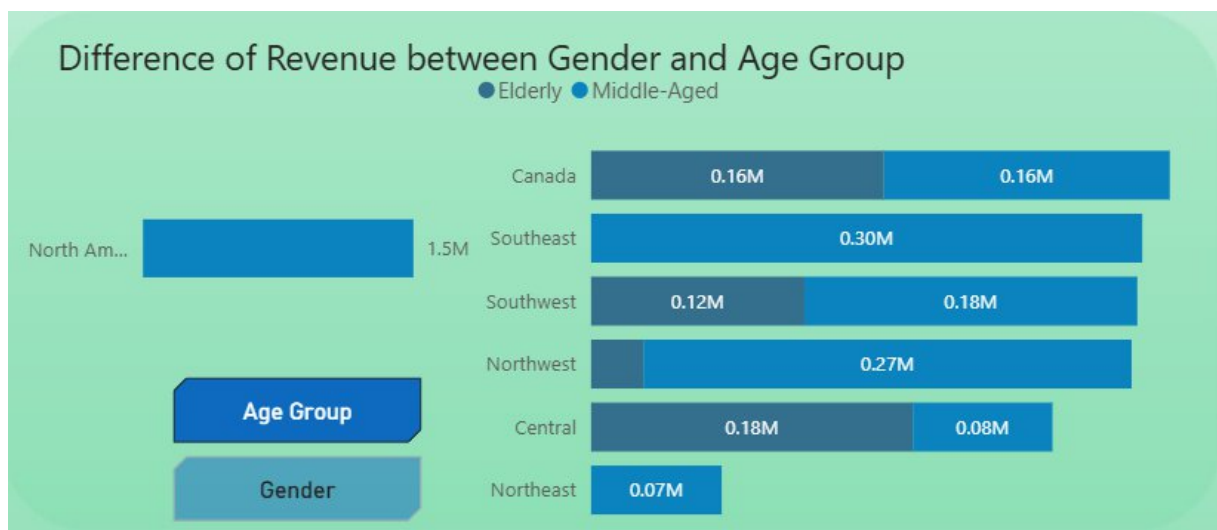


Figure 3.15. Employee analysis by age group in January 2010

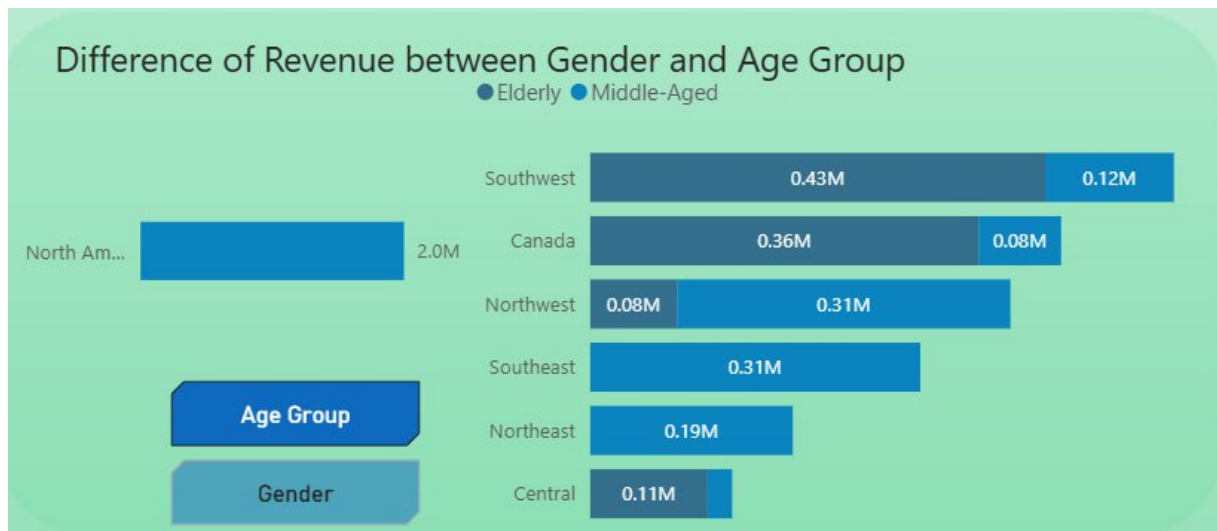


Figure 3.16. Employee analysis by age group in March 2010

For this "pioneering" strategy to succeed, experienced personnel were the core element. In January and March 2010, it is easy to see that reseller revenue primarily came from the experienced sales group aged 45 and above, particularly the Middle-Aged group (45-60 years old), contributing about 58.62% of total revenue in January and March 2010. This group possessed the necessary experience while retaining the agility and sharpness to operate in the open market. In the initial launch months when the AdventureWorks brand was still new, convincing resellers to accept the risk of large-scale inventory purchases depended entirely on the personal credibility and existing relationships of the veteran team. However, an interesting shift occurred in May 2010 when employees from younger age groups began to breakthrough in sales. This suggests the sales process was gradually being standardized, and the brand had started to establish its own position, enabling employees to close deals based on product effectiveness rather than solely on interpersonal relationships.

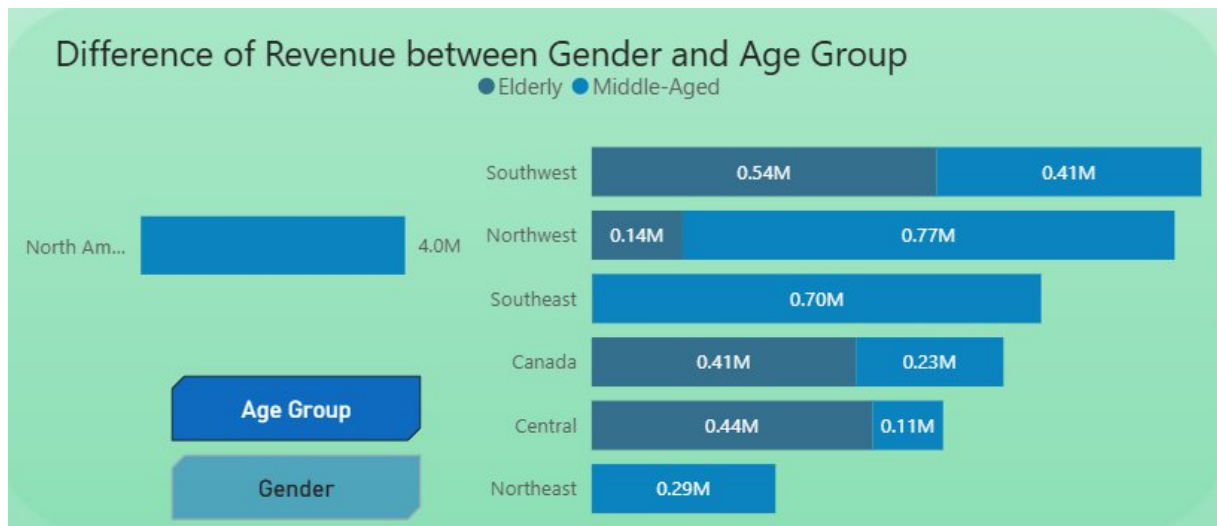


Figure 3.17. Employee analysis by age group in May 2010

The strength of the workforce was further reflected in the complementary roles of gender in negotiation and market maintenance. In the early months of the year, the male sales team dominated absolutely, leveraging their ability to secure "massive" contracts for market coverage. However, by May, sales from accounts managed by female employees showed more sustainable growth. This insight reveals that while men excelled in "capturing" new territories, women demonstrated superior strengths in customer care and motivating resellers to re-order as the peak season approached, ensuring stable cash flow for the business.

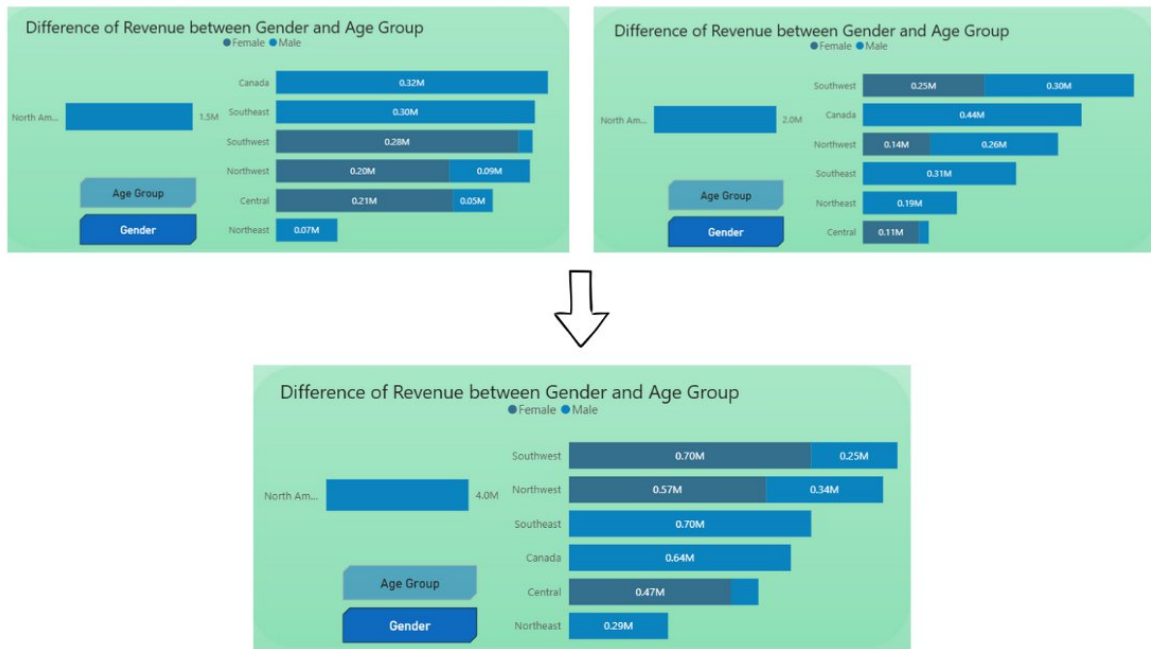


Figure 3.18. Employee analysis by Gender in January, March and May 2010

Finally, the tight linkage between products and sales timing completed the strategic picture for 2010. Nearly 100% of Reseller revenue was still concentrated on Bikes, but with a smart diversification of types. While the Internet channel sold Road Bikes effectively, resellers began importing large quantities of Mountain Bikes. This was a strategic inventory preparation for the summer travel season, explaining why Internet Sales exploded in subsequent months. Thus, the Reseller channel acted as a "scale lever" to blanket the market, while Internet Sales served as a sophisticated "sensor," helping AdventureWorks capture consumer trends, creating a complementary multi-channel sales ecosystem from the early days of establishment.

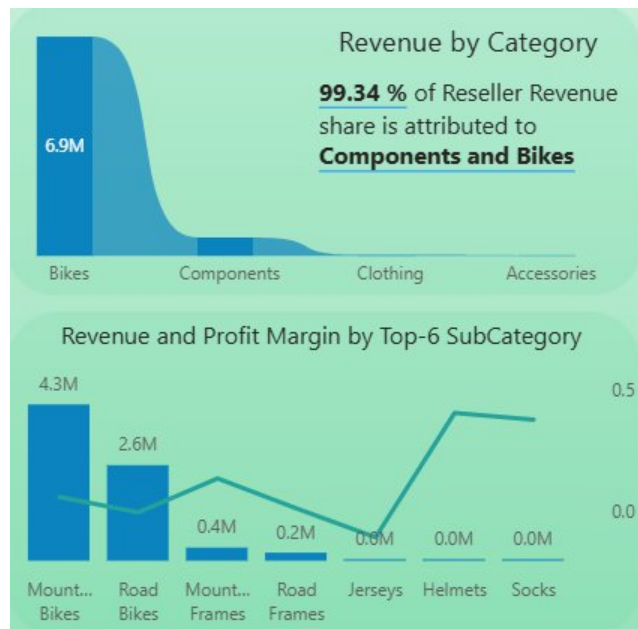


Figure 3.19. Revenue analysis by product of Reseller Sales in 2010

Moving to 2011, this year marked a strong transformation for AdventureWorks as the Reseller channel moved away from the batch sales model to continuous 12-month supply. Revenue in 2011 grew remarkably compared to 2010, with total annual revenue reaching \$25.95 million USD, nearly 3.5 times the previous year, peaking in Q3 with approximately \$8.3 million in quarterly revenue. This result indicates the "Push Strategy" operated smoothly, with resellers proactively stocking up on large-scale inventory in anticipation of the holiday season. Notably, although Bikes still accounted for nearly 100% of revenue share, the product structure shifted remarkably from Mountain Bikes leading revenue in 2010 to Road Bikes—mirroring the Internet channel—with \$11.7 million in revenue, corresponding to about 45.1% of total 2011 revenue.

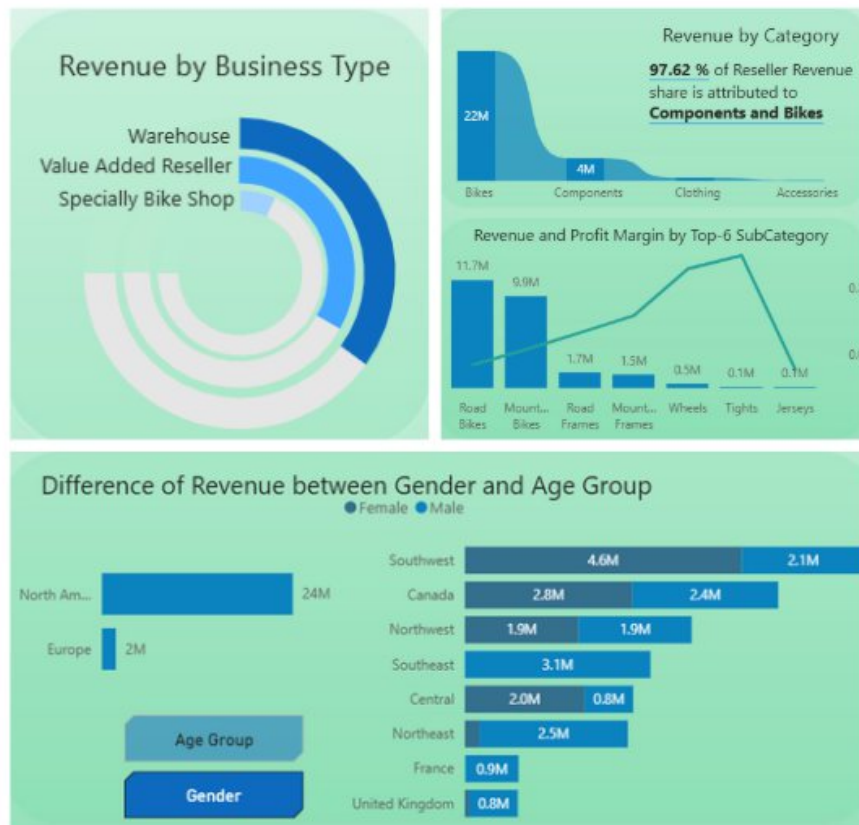


Figure 3.20. Revenue analysis of Reseller Sales in 2011

The strength of the wholesale channel in 2011 was further illustrated through a "divide and conquer" strategy across different geographical "battlefields" and business types. In North America, Warehouse and Value Added Resellers acted as the "revenue backbone," prioritizing massive shipments of mass-market bike lines. Meanwhile, in European markets like France and the UK, inventory was mostly distributed to Warehouse-type resellers.

However, to access such a large number of resellers, the contribution of the account management team was crucial. Overall, most of the 2011 revenue still came from employees aged 45 to 60, with a relatively balanced gender mix. This combination of employee groups created a "safety buffer zone," helping maintain stable company cash flow even during low-season months.

Another valuable insight emerged from cross-referencing the two main sales channels: the "Buffer Inventory" effect. Data shows that from mid-2011, Resellers began to increase their purchases of Mountain Bikes, even though consumer demand on the Internet channel had not yet exploded at that time. This proves that AdventureWorks used the reseller network as a satellite warehouse system to alleviate central inventory pressure and minimize logistics risks. Pushing inventory to Resellers early not only optimized operations for the parent company but also ensured goods were readily available locally, creating the most favorable conditions to immediately meet consumer demand when they started searching for products on the online platform.

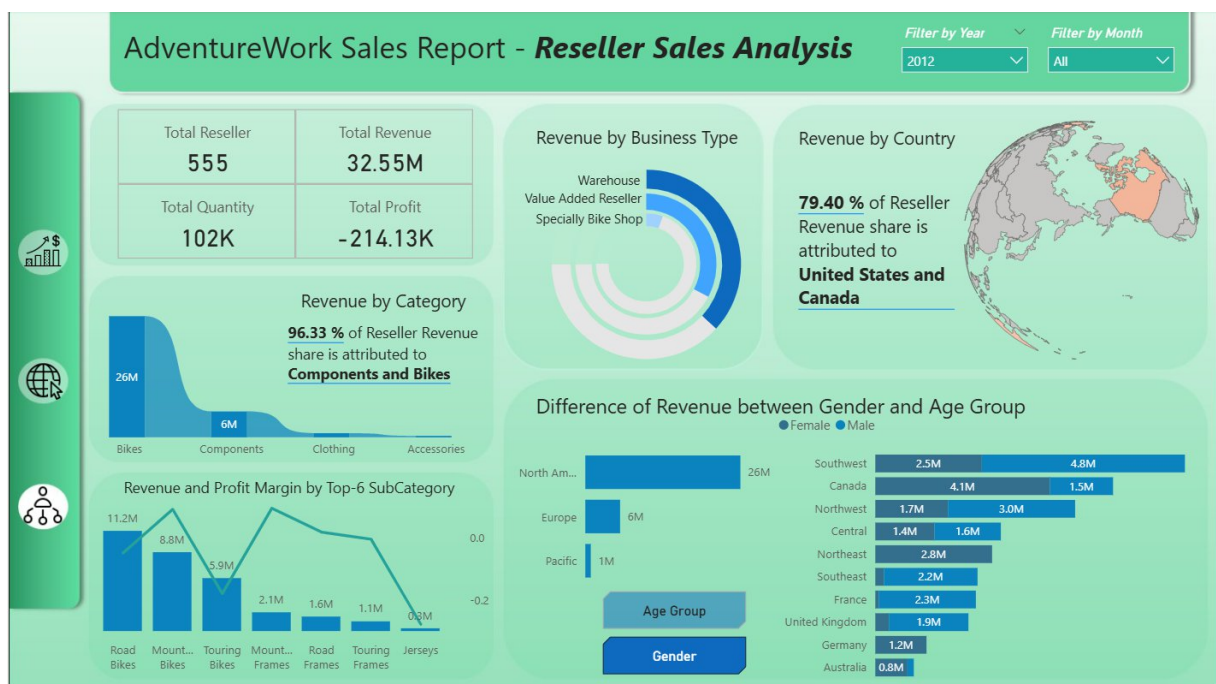


Figure 3.21. Revenue analysis of Reseller Sales in 2012

By 2012, the Reseller channel entered a phase of "stable saturation" following the previous year's meteoric growth cycle. Revenue this year no longer saw leaps but began to plateau, partly as a consequence of resellers having imported excessively large quantities in late 2011 and spending the first half of 2012 focusing on clearing that inventory. Although some months still recorded high sales, the growth rate slowed significantly, indicating that Resellers had reached scale limits in traditional markets. Growth at this point no longer came from expanding the number of resellers but

depended entirely on optimizing the product portfolio to adapt to the rapidly changing market context.

Particularly, due to reseller resistance to new inventory stemming from overstock, the company implemented various promotional programs for resellers. However, this increased product costs and led to consecutive negative profits in January (loss >\$250,000), February (loss >\$270,000), and March (loss >\$3,000). Consequently, the total annual profit for 2012 was also negative, with a loss exceeding \$200,000. Regarding business regions, the primary revenue for the Reseller channel came from North America, with \$26 million in revenue, accounting for approximately 79.88% of 2012 revenue.

In contrast to the Reseller channel, 2012 was a year of brilliant growth for the online channel. This inadvertently created a "Channel Conflict" effect, exerting reverse pressure on the wholesale system. When retail customers flocked to the website to buy cheap and diverse accessories, Resellers who traditionally focused on selling complete bikes began to feel threatened. The consequence was that sales of mass-market bike lines through Resellers were affected, as customers could easily compare prices online. Resellers began showing signs of resistance by reducing orders for mass-market models and focusing only on high-end segments requiring direct consultation. This was a warning signal forcing the company to rebalance benefits between distribution channels to avoid unnecessary internal competition.

The financial picture of the last two months of 2012 further reinforced these strategic warnings. Although the number of shipments (Total Quantity) increased, actual profits were negative. The main reason stemmed from AdventureWorks having to offer stronger discount policies to Warehouse-type resellers to maintain annual sales targets. This insight shows that using discounts is a double-edged sword: it can protect short-term sales but is quietly eroding net profit margins, posing a significant challenge for maintaining sustainable financial health in the upcoming year of 2013.

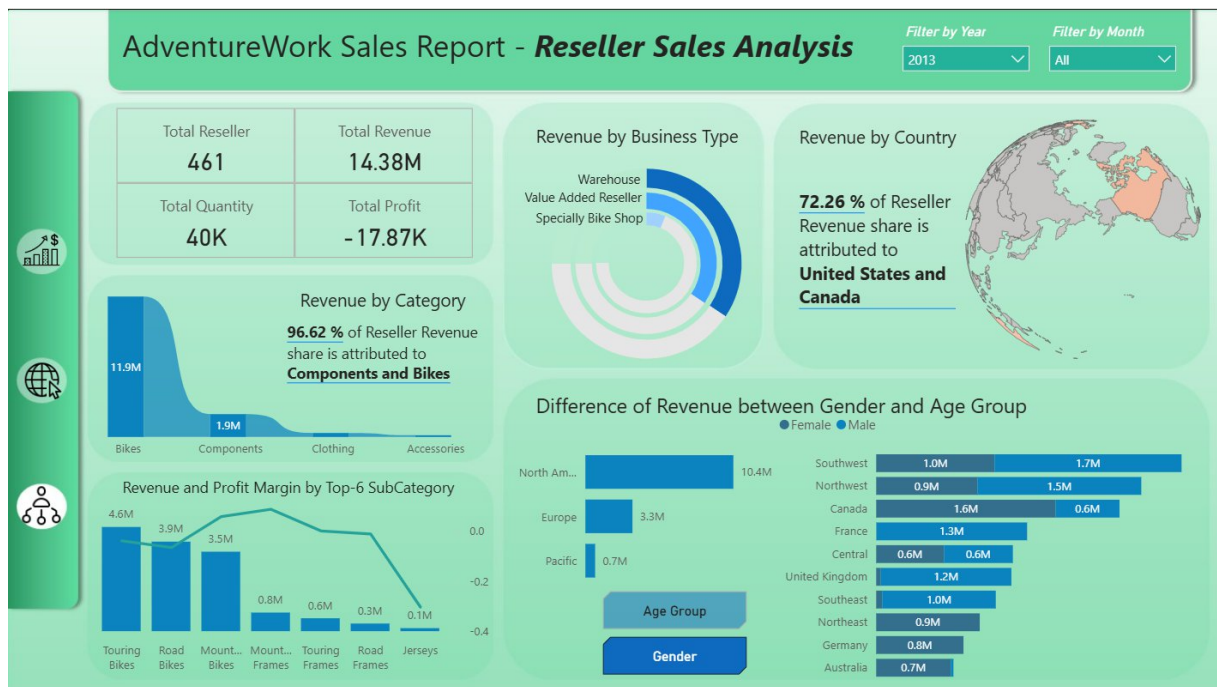


Figure 3.22. Revenue analysis of Reseller Sales in 2013

Moving to the final year of the 2010–2013 period, 2013 recorded a highly volatile scenario for AdventureWorks' Reseller channel, as sales data completely "disappeared" in the first half of the year and again in the final month of December. The data void from January to June holds a different meaning compared to the Internet channel; it is likely a consequence of a system freeze to facilitate a comprehensive ERP upgrade. Given that the wholesale channel possesses a significantly more complex operational process—from managing accounts payable to multi-tiered reseller discount structures—the system transition caused a complete standstill in recording new orders. Particularly, the lack of data for December is an extremely dangerous signal, indicating that after five months of efforts to recover (July–November), the enterprise may have encountered serious supply chain issues or proactively halted operations for inventory assessment and restructuring in the face of excessively large losses.

Furthermore, the decline of Profit into negative territory during this period (-\$17,870 USD) is the result of three "fatal blows" to the company's financial health. The first is Liquidation Costs. To pave the way for new 2014 bike models, AdventureWorks was forced to push all remaining old inventory from 2011–2012 to its Resellers. To

persuade resellers to accept these outdated batches, the company had to implement a "fire sale" discount campaign. At this point, the actual selling price was much lower than the production cost, leading to a situation where the more they sold, the deeper the losses, as revenue was insufficient to cover the cost of goods sold.

Running parallel to the inventory clearance pressure was AdventureWorks' weakened position in negotiations with large Warehouses, particularly in the US market. Amid difficult economic conditions, these partners leveraged their high bargaining power to force the company to bear all shipping and warehousing costs. The consequence was that although reports still recorded revenue, after deducting variable costs and excessive trade discounts, the net profit of the wholesale channel was eroded and plunged directly into negative territory. This is clear evidence that the company was forced to "buy" revenue at any cost to maintain cash flow.

2.4. Insight from sale analysis

Thus, through the analysis of AdventureWorks' sales data for the 2010-2013 period, it is evident that this was a journey of dynamic transformation from a foundation-establishing enterprise to a complex multi-channel business entity. The year 2010 is regarded as a pivotal milestone, where the company began shaping a data-driven management philosophy, moving away from intuition. With an initial modest total revenue of \$10.8 million USD, AdventureWorks wisely chose a "Dual-channel" model. Within this, the wholesale (Reseller) channel in North America acted as the brand "anchor," enabling rapid market penetration and capturing over 70% of revenue in the first year. The strategic acceptance of lower profit margins during this phase was a deliberate sacrifice in exchange for a deep understanding of logistics cost structures and Cost of Goods Sold (COGS), laying a solid foundation for the explosive growth in subsequent years.

Based on the analytical dashboards, we can extract critical "insights" vital to AdventureWorks' business model. The first is the phenomenon of "interconnected vessels" between the two distribution channels. The company utilized the Internet Sales

channel—characterized by high profitability and good stability—as a "safety net" to compensate for the extreme volatility of the Reseller channel. Second, the "Buffer Inventory" strategy was implemented in a sophisticated manner; resellers were not merely sales points but also served as satellite warehouses, helping AdventureWorks alleviate inventory pressure at central hubs and minimize logistics risks during peak seasons.

Furthermore, another crucial insight stems from the shift towards a "Long-tail Strategy" starting in 2013. Instead of solely focusing on selling new, high-value bicycles with low purchase frequency, the company began to maximize "Customer Lifetime Value" (LTV) through the retail of components, accessories, and specialized apparel. This not only helped maintain stable cash flow but also fostered longer-term customer retention within the brand's ecosystem.

A deep dive into customer profiling reveals clear demographic segmentation aligned with each product line. Bicycle buyers are primarily high-income individuals (over \$80,000 USD) aged 49 to 69. Meanwhile, accessory buyers are concentrated in an older age group (65-82), who are particularly concerned with health and safety. Conversely, the apparel segment attracts younger customers (under 45), who value aesthetics and sports fashion trends.

Additionally, a unique point in the report is the influence of gender on the business performance of the sales team. Data indicates that men often have a distinct advantage in exploring new markets and securing large-scale reseller contracts. However, the female sales team demonstrates greater resilience and effectiveness in customer care and motivating resellers to re-order. Leveraging these respective strengths of the personnel contributed significantly to the stable sales growth during the 2011-2012 period.

The greatest strength of AdventureWorks during this phase was its ability to optimize profit margins through the Internet channel. With an extremely low Cost of Goods Sold (approximately 21% of the company's total COGS in 2010), the Internet

channel became the primary profit engine. By 2012, the company achieved Economies of Scale, reducing marginal costs and creating the widest profit gap in the four-year history.

Geographic diversification capability was also a highlight, particularly the explosive growth of the Australian market via the Internet channel. This market served as a counterweight, helping offset cyclical fluctuations in traditional markets like the US and Europe. Moreover, understanding the operational rhythm of the logistics system and identifying the "golden days" of the week for processing wholesale and retail orders enabled the company to save significantly on operational costs and improve on-time order fulfillment rates.

However, alongside these brilliant successes, AdventureWorks' operational system also exposed critical vulnerabilities. Over-reliance on the reseller channel caused the company's profits to fluctuate in a "sawtooth" pattern, making them extremely sensitive to market volatility. The logistics system, despite improvements, still encountered "bottleneck" issues in 2012 when the explosion of small retail orders for accessories and apparel exceeded warehouse processing capacity, forcing the company to temporarily halt sales of these items to prioritize core bicycle lines.

Particularly, the phase shift in data between Quantity, Revenue, and Gross Profit in late 2011 served as a warning signal about ineffective cost control. While revenue could grow, without tight management of COGS and variable operational costs, actual realized profits would be quickly eroded.

The most serious issue emerged in 2013, when revenue from the Reseller channel nearly disappeared entirely in the first half of the year without clear reasons in preliminary reports. This incident, combined with inventory shortages following large-scale clearance campaigns in late 2012, fractured the supply chain.

The report identifies "three fatal blows" to the company's financial health in this final phase. The first was excessively high Liquidation Costs to pave the way for 2014 bike models. The second was a "fire sale" discount campaign that drove actual selling

prices below production costs, leading to a situation where more sales resulted in deeper losses (a negative profit exceeding \$17,000 USD). The third was AdventureWorks' weakened negotiating position with large US wholesalers, forcing the company to bear all shipping and warehousing costs, completely erasing the net profit of the wholesale channel.

Therefore, to optimize future operations and avoid repeating past mistakes, AdventureWorks must answer key strategic questions.

First, the company needs to determine whether the previous period's revenue explosion was driven by genuine market demand or merely a consequence of short-term promotional programs.

Second, how can a more accurate inventory forecasting mechanism be built to prevent "stockouts" when market demand recovers?

Third, the company needs to restructure its bargaining power with reseller and wholesale partners. Should we continue maintaining the current Reseller model, or is a stronger shift towards a Direct-to-Consumer (D2C) model necessary?

Finally, decoding the misalignment between financial and operational metrics is an urgent task to ensure that all revenue growth efforts lead to a corresponding increase in actual profit.

3. Recommendation for AdventureWork

Based on the analysis and insights uncovered through this process, it is evident that AdventureWorks stands at a critical crossroads. The successes from market penetration and optimizing the Internet channel provide a solid foundation. However, without thoroughly resolving issues related to the supply chain, liquidation cost control, and the stability of the reseller channel, the enterprise will struggle to sustain its development momentum in subsequent phases. A strategy that balances scale expansion with profit margin control is the key for AdventureWorks to regain its position in the

global bicycle market. To achieve this, this analysis aims to provide some recommendations tailored for AdventureWorks.

First, the most important recommendation lies in restructuring the distribution channel system. Insights from the data reveal that the Reseller channel, while generating massive sales, is the source of profit instability and inventory pressure. AdventureWorks needs to shift from a "Push strategy" model to a "Pull strategy" model through an integrated Omni-channel strategy. Instead of viewing resellers as dumping grounds for old inventory, the enterprise should professionalize this relationship by implementing a Reseller Tiering system. Diamond or Gold-tier resellers must be committed to new product roadmaps and receive marketing support, but in return, they must share actual sell-through data with AdventureWorks. This data transparency from resellers will help the company avoid "information blindness," preventing overproduction or loss-making inventory clearances like the one in 2013.

Next, concurrently with rectifying the wholesale channel, AdventureWorks must intensify investment in the Direct-to-Consumer (D2C) channel via the Internet Sales platform. This is the channel with ideal profit margins and where the enterprise directly grasps the market's "pulse." Optimizing the user experience on the website and integrating tools for bike selection based on body type and usage needs will transform the Internet channel from just a sales point into a customer data funnel. By holding detailed customer profiles, AdventureWorks can execute personalized remarketing campaigns, thereby reducing Customer Acquisition Cost (CAC) and increasing customer retention rates.

Third, the stockout issue at the beginning of 2013 following a loss-making clearance campaign demonstrates a deficiency in Product Lifecycle Management (PLM). Therefore, the next recommendation is for AdventureWorks to establish a scientific Phased Rollout process. Instead of concentrating all pressure to clear old inventory in the last quarter of the year, the company should apply a Dynamic Markdown strategy based on the inventory age of specific product lines. For example, Mountain Bikes, which become obsolete faster than Road Bikes, need demand

stimulation earlier through trade-in programs. This not only helps maintain stable cash flow but also preserves brand value from being devalued by "fire sale" campaigns.

Furthermore, the logistics system needs upgrading to handle the diversification of order types. The 2012 "bottleneck" incident when accessory orders surged shows the company was unprepared for a retail shipping model. AdventureWorks should consider segregating warehouse processes: one department dedicated to packaging and shipping large pallet shipments for Resellers, and another highly automated department for small retail orders (e-fulfillment). Utilizing third-party logistics (3PL) providers in distant markets like Australia or Europe is also a necessary option to reduce direct management burdens while ensuring competitive delivery lead times.

Fourth, a highlight from the analysis is that AdventureWorks already has a fairly clear picture of its customers by age and income. The strategic recommendation here is to transition from Mass Marketing to Segmented Marketing. For the core customer group of high-income individuals aged 49-69, communication campaigns should not only focus on technical specifications but also emphasize "lifestyle" and "conquering experiences." Family Bundling products would be an excellent lever. For instance, when a father buys a premium bike, the company could offer discounted bundles of protective gear for children or sportswear for the spouse. This taps into the family care psychology of this group while increasing the Average Order Value (AOV).

For the younger customer group buying Clothing, the strategy should focus on fashion and community connection. Creating virtual cycling clubs on apps or social media where users can share routes and showcase outfits will generate a natural Viral effect. This is also a way for AdventureWorks to collect data on color and style trends favored by young people, providing feedback to the product design department to minimize the risk of holding unfashionable inventory.

Fifth, personnel are often overlooked in financial reports, but AdventureWorks' data revealed clear performance differences based on gender and work nature. The management recommendation is for the company to apply a "Hunters & Farmers"

model. The male team, with strengths in exploration, should be tasked with expanding the reseller network into new territories or negotiating large supply contracts for public projects. Conversely, the female team, with their attention to detail and strong relationship-building skills, should focus on Customer Relationship Management (CRM) and reseller support. Optimizing personnel according to their strengths will not only increase productivity but also reduce turnover, helping the company retain experienced staff.

Regarding operational processes, AdventureWorks needs to leverage the insight about "golden days" of the week. Concentrating efforts on processing wholesale orders mid-week and retail orders on other days will help balance the load on the warehouse system. The company should also consider implementing flexible work schedules or hiring seasonal staff during peak hours identified from historical data to optimize labor costs per order.

Finally, to decisively resolve the misalignment between revenue and profit, AdventureWorks needs an Early Warning System for finance. Any discount decisions or promotional programs from the sales department must have their impact on gross profit simulated before approval. The situation of "selling more, losing more" is a consequence of the lack of coordination between the Sales and Finance departments.

In particular, in large markets like the US, the enterprise must proactively seek ways to balance negotiating power with wholesale partners. Diversifying the list of warehousing and shipping service providers will prevent AdventureWorks from being passive when partners pressure storage costs. Additionally, investing in smaller, centrally-located distribution centers near key consumption areas will help reduce Last-mile delivery costs – one of the most expensive costs in e-commerce.

REFERENCE

From Dataset provided by the instructor

